8.[Page Replacement Policy]

A system has RAM which can store four pages simultaneously satisfy page requests. Write a Program to calculate percentage pages hit and page fault if the system uses Least recently used page replacement, technique for the page references entered by the user i.e. 1, 2, 3, 4, 1, 2, 5, 1, 2, 3, 4, 5 (No of Frames=3)

CODE:

```
GNU nano 8.7                         lru.c                          Modified
nclude<stdio.h>

t main()

   int pages[] = {1,2,3,4,1,2,5,1,2,3,4,5};
   int frames[3];
   int i,j,k,found=0;
   int page_fault=0,page_hit=0;
   int count=12;

   for(i=0;i<3;i++)
       frames[i]=-1;

   for(i=0;i<count;i++)
   {
       found=0;

       for(j=0;j<3;j++)
       {
           if(frames[j]==pages[i])
           {
               found=1;
               page_hit++;
               break;
           }
       }

       if(found==0)
       {
           frames[i%3]=pages[i];
           page_fault++;
       }

       printf("\nFrames: ");
       for(k=0;k<3;k++)
           printf("%d ",frames[k]);
   }

   printf("\n\nTotal Page Faults = %d",page_fault);
   printf("\nTotal Page Hits = %d",page_hit);

   printf("\nPage Fault Percentage = %.2f",(page_fault/(float)count)*100);
   printf("\nPage Hit Percentage = %.2f",(page_hit/(float)count)*100);

 Help       ^O Write Out  ^F Where Is   ^K Cut        ^T Execute    ^C Location   M-U Undo
 Exit       ^R Read File  ^\ Replace    ^U Paste      ^J Justify    ^/ Go To Line M-E Redo
```

```
GNU nano 8.7                         lru.c                         Modified
    int pages[] = {1,2,3,4,1,2,5,1,2,3,4,5};
    int frames[3];
    int i,j,k,found=0;
    int page_fault=0,page_hit=0;
    int count=12;

    for(i=0;i<3;i++)
        frames[i]=-1;

    for(i=0;i<count;i++)
    {
        found=0;

        for(j=0;j<3;j++)
        {
            if(frames[j]==pages[i])
            {
                found=1;
                page_hit++;
                break;
            }
        }

        if(found==0)
        {
            frames[i%3]=pages[i];
            page_fault++;
        }

        printf("\nFrames: ");
        for(k=0;k<3;k++)
            printf("%d ",frames[k]);
    }

    printf("\n\nTotal Page Faults = %d",page_fault);
    printf("\nTotal Page Hits = %d",page_hit);

    printf("\nPage Fault Percentage = %.2f",(page_fault/(float)count)*100);
    printf("\nPage Hit Percentage = %.2f",(page_hit/(float)count)*100);

    return 0;

^G Help      ^O Write Out  ^F Where Is  ^K Cut     ^T Execute  ^C Location    M-U Undo
^X Exit      ^R Read File  ^\ Replace   ^U Paste   ^J Justify  ^/ Go To Line  M-E Redo
```

OUTPUT:

```
ASUS@Kalash-Laptop MINGW64 ~
$ ^[[200~cd Desktop~
-bash: $'\E[200~cd': command not found

ASUS@Kalash-Laptop MINGW64 ~
$ cd Desktop
-bash: cd: Desktop: No such file or directory

ASUS@Kalash-Laptop MINGW64 ~
$ nano lru.c

ASUS@Kalash-Laptop MINGW64 ~
$ gcc lru.c -o lru

ASUS@Kalash-Laptop MINGW64 ~
$ ./lru

Frames: 1 -1 -1
Frames: 1 2 -1
Frames: 1 2 3
Frames: 4 2 3
Frames: 4 1 3
Frames: 4 1 2
Frames: 5 1 2
Frames: 5 1 2
Frames: 5 1 2
Frames: 3 1 2
Frames: 3 4 2
Frames: 3 4 5

Total Page Faults = 10
Total Page Hits = 2
Page Fault Percentage = 83.33
Page Hit Percentage = 16.67
ASUS@Kalash-Laptop MINGW64 ~
$
```